Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 1.9

1. Assume that T is a linear transformation. Find the standard matrix of T.

   T: $\mathbb{R}^2 \to \mathbb{R}^4$, $T(\mathbf{e}_1) = (4, 1, 4, 1)$, and $T(\mathbf{e}_2) = (-9, 3, 0, 0)$, where $\mathbf{e}_1 = (1,0)$ and $\mathbf{e}_2 = (0,1)$.

   $A = \begin{bmatrix} 4 & -9 \\ 1 & 3 \\ 4 & 0 \\ 1 & 0 \end{bmatrix}$ (Type an integer or decimal for each matrix element.)

2. Assume that T is a linear transformation. Find the standard matrix of T.

   T: $\mathbb{R}^3 \to \mathbb{R}^2$, $T(\mathbf{e}_1) = (1,9)$, and $T(\mathbf{e}_2) = (-3,2)$, and $T(\mathbf{e}_3) = (9, -8)$, where $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are the columns of the 3×3 identity matrix.

   $A = \begin{bmatrix} 1 & -3 & 9 \\ 9 & 2 & -8 \end{bmatrix}$ (Type an integer or decimal for each matrix element.)

3. Assume that T is a linear transformation. Find the standard matrix of T.

   T: $\mathbb{R}^2 \to \mathbb{R}^2$, rotates points (about the origin) through $-\frac{2\pi}{3}$ radians.

   $A = \begin{bmatrix} \cos\left(-\frac{2\pi}{3}\right) & -\sin\left(-\frac{2\pi}{3}\right) \\ \sin\left(-\frac{2\pi}{3}\right) & \cos\left(-\frac{2\pi}{3}\right) \end{bmatrix}$

   (Type an integer or simplified fraction for each matrix element. Type exact answers, using radicals as needed.)

4. Assume that T is a linear transformation. Find the standard matrix of T.

   T: $\mathbb{R}^2 \to \mathbb{R}^2$ is a vertical shear transformation that maps $\mathbf{e}_1$ into $\mathbf{e}_1 - 8\mathbf{e}_2$ but leaves the vector $\mathbf{e}_2$ unchanged.

   $A = \begin{bmatrix} 1 & 0 \\ -8 & 1 \end{bmatrix}$

   (Type an integer or simplified fraction for each matrix element.)

5. Assume that T is a linear transformation. Find the standard matrix of T.

   T: $\mathbb{R}^2 \to \mathbb{R}^2$ first reflects points through the horizontal x_1-axis and then reflects points through the vertical x_2-axis.

   $A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

   (Type an integer or simplified fraction for each matrix element.)

6. Assume that T is a linear transformation. Find the standard matrix of T.

   T: $\mathbb{R}^2 \to \mathbb{R}^2$, first performs a horizontal shear that transforms $\mathbf{e}_2$ into $\mathbf{e}_2 + 3\mathbf{e}_1$ (leaving $\mathbf{e}_1$ unchanged) and then reflects points through the line $x_2 = -x_1$.

   $A = \begin{bmatrix} 0 & -1 \\ -1 & -3 \end{bmatrix}$

   (Type an integer or simplified fraction for each matrix element.)

7. Let T: $\mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x_1, x_2) = (x_1 + x_2, 4x_1 + 6x_2)$. Find **x** such that T(**x**) = (3,4).

x = $\begin{bmatrix} 7 \\ -4 \end{bmatrix}$

8. Determine whether the statement below is true or false. Justify the answer.

A linear transformation T: $\mathbb{R}^n \to \mathbb{R}^m$ is completely determined by its effect on the columns of the $n \times n$ identity matrix.

Choose the correct answer below.

- **A.** The statement is true. A transformation T is defined in terms of its effect on the columns of the $n \times n$ identity matrix because otherwise the standard matrix A would be impossible to determine.
- **B.** The statement is false. It is possible for more than one linear transformation to have the same effect on the columns of the $n \times n$ identity matrix.
- **C.** The statement is true. The vector **x** can be written as a linear combination of the columns of the identity matrix. T is a linear transformation so T(**x**) can be written as a linear combination of the vectors T($\mathbf{e}_1$) through T($\mathbf{e}_n$).
- **D.** The statement is false. The transformation is jointly determined by its effect on the columns of the $n \times n$ identity matrix and by the result of writing **x** as a linear combination of the columns of the standard matrix A.

9. Determine whether the statement below is true or false. Justify the answer.

If T: $\mathbb{R}^2 \to \mathbb{R}^2$ rotates vectors about the origin through an angle φ, then T is a linear transformation.

Choose the correct answer below.

- **A.** The statement is true. The standard matrix, A, of the linear transformation is $\begin{bmatrix} \sin\varphi & \cos\varphi \\ -\cos\varphi & \sin\varphi \end{bmatrix}$.
- **B.** The statement is false. The standard matrix, A, of the linear transformation is $\begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & -\cos\varphi \end{bmatrix}$, but because the sine and cosine functions are nonlinear, the transformation is not linear.
- **C.** The statement is true. The standard matrix, A, of the linear transformation is $\begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$.
- **D.** The statement is false. Rotations move points around a circle and so are not linear transformations.

10. Determine whether the statement below is true or false. Justify the answer.

The columns of the standard matrix for a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ are the images of the columns of the $n \times n$ identity matrix.

Choose the correct answer below.

- **A.** The statement is false. The standard matrix only has the trivial solution.
- **B.** The statement is false. The standard matrix is the $m \times n$ matrix whose jth column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the jth column whose entries are all 0.
- **C.** The statement is true. The standard matrix is the identity matrix in $\mathbb{R}^n$.
- **D.** The statement is true. The standard matrix is the $m \times n$ matrix whose jth column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the jth column of the identity matrix in $\mathbb{R}^n$.

11. Determine whether the statement below is true or false. Justify the answer.

A mapping T: $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is onto $\mathbb{R}^m$ if every vector **x** in $\mathbb{R}^n$ maps onto some vector in $\mathbb{R}^m$.

Choose the correct answer below.

- **A.** The statement is false. A mapping T: $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is one-to-one if every vector **x** in $\mathbb{R}^n$ maps onto some vector in $\mathbb{R}^m$.
- **B.** The statement is false. A mapping T: $\mathbb{R}^n \rightarrow \mathbb{R}^m$ is onto $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is mapped onto by some vector **x** in $\mathbb{R}^n$.
- **C.** The statement is true. A transformation is onto when, for every **b** in the codomain, the matrix equation A**x** = **b** has a unique solution.
- **D.** The statement is true. A transformation is onto when all possible values in the codomain, or range, are mapped to by some value in the domain.